PROJECT PROPOSAL

Date of proposal : 03/10/2025
Project Title : Data-Driven Football game Outcome Forecasting with Managerial Insights
Group ID (As Enrolled in Canvas Class Groups) : Group 3 **Group Members (name , Student ID) :** Zhang Hengda (A0329373X) Sandesh Sreepathy Upadhyaya (A0327834X) Shantam Sharma (A0329623B)
Sponsor/Client: *(Company Name, Address and Contact Name, Email, if any) : NA*
Background/Aims/Objectives: We aim to design and implement a system that predicts the final scoreline of the football match and provides actionable feedback for team managers to improve decision-making and strategy.
Project Descriptions: This project aims to build an AI-powered system that predicts the scoreline of a match based on the current form of players and teams. The solution can optionally suggest a best team lineup (like Dream 11) by analyzing player performance data, fitness, and recent statistics. Beyond prediction, the system will generate actionable feedback for the team managers, recommending strategies such as optimal player rotations, substitutions, and formation adjustments to improve match outcomes. **Business Use Case:** The solution will act as a decision-support tool for sports managers, providing: • Scoreline Predictions: Anticipating end-of-match outcomes and analyze key contributors for the same • Managerial Feedback: Offering practical recommendations on player selection, player performance improvement and overall team-related strategies. By leveraging predictive analytics and machine learning, this project helps football team managers make informed, evidence-based decisions, bolstering visibility into team performance, player's current form and match specific strategies. **Dataset Overview:** Below are the datasets which will be used for the project: 1. Football Statistics and History | FBref.com 2. Football Player Statistics EURO/Euro Leagues 3. Football Players Data

The project leverages a structured dataset encompassing approximately 20 English Premier League teams, with detailed player-level statistics. The key elements of the dataset include:

- **Team Information:** Each team's identity, recent performance trends, and league standings.
- **Player Statistics:** Metrics such as goals, assists, pass accuracy, tackles, shots on target and fitness levels, which will be utilized to track player form. This also enables evaluation of individual contributions and performance patterns.
- **Match Context:** Historical match outcomes, scorelines, and situational factors that influence game results.

We have also found open-source websites that allow web-scraping or have a free API to extract data for current EPL season, which can be utilized to test and validate our models.

Techniques And Methodologies:

Below are some techniques discussed in ISY-5001 which we aim to implement to complete our deliverables.

Clustering with K-Means:

- Used to identify distinct player styles and roles based on their statistical attributes (e.g., attacking vs. defensive tendencies, playmaking efficiency, pressing intensity).
- Helps in analyzing the current form of individual players by grouping them into performance-based clusters.

Predictive Modeling with Non-Linear Regression Approaches:

- Techniques such as Neural Networks (NNs) and Random Forests will be applied to predict the probable match outcomes and scorelines.
- The models will take into account the combination of player styles across both starting lineups (11 vs 11) to simulate how the teams are likely to perform against each other.

Prompt-Engineering

- Generate insights and recommendations using LLM based on knowledge derived from trained models and clustering techniques

Final Deliverables:

Since we will maintain a database for player and team performance, we aim to deliver following as our final product:

- Chatbot UI to get information on player/team performance
- Upcoming match score-line prediction with model-based reasoning
- Insights into player statistics and current playing form
- Create best 11-player football team across entire league which can beat any other combination of players
- Recommendations for each team manager to improve team/player performance